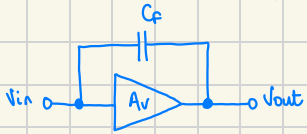
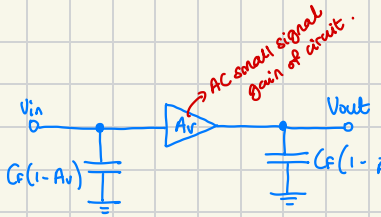
EE& 3136

Lecture 4

Miller Multiplication for capacitors



->



$A_V = \frac{V_{out}}{V_{in}}$

Finding poles using pole identification

Step 1: Disable all DC sources

Step 2: Spot all capacitors

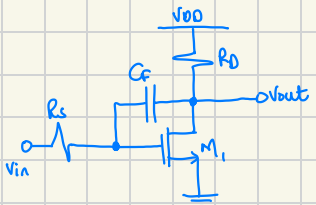
Step 3: Deal with capacitors connected to GND at one node first.

Step 4: Apply Miller's theorem for ungrounded capacitors.

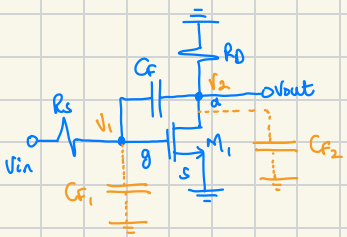
Step 5: All capacitors that are parallel to a resistor is going produce a pole.

Vin is also disabled during pole identification since we disable all independent sources when doing Thevenin.

Example:



=>



$C_{f_1} = (1 - A_V)\, C_f$

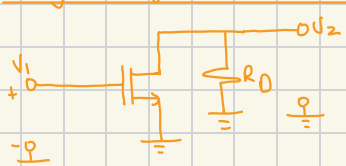
$C_{f_2} = (1 - \frac{1}{A_V})\, C_f$

$A_V = ?$

Input given in gate, output taken from drain => "common-source" common source gain = $-g_m R_{out}$, $g_m = g_m$

$= -g_m R_D$

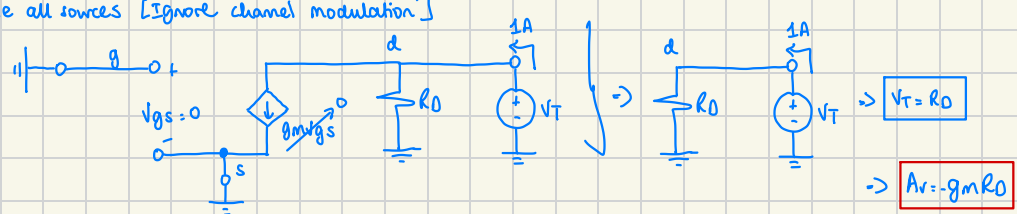
Finding Av if you don't remember the gain of common-source [Ignore capacitors when computing Av]



Finding G_m: ? BRO CODE says: $G_M = g_m$

Finding Rout

Disable all sources [Ignore channel modulation]

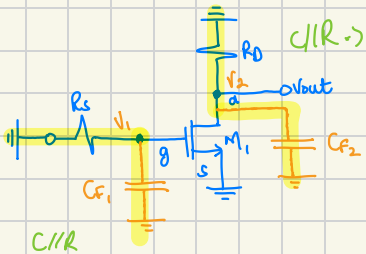


=> $V_T = R_D$

=> $A_V = -g_m R_D$

=> $C_{f_1} = (1 + g_m R_D)\, C_f = (1 + g_m R_D)\, C_f$

$C_{f_2} = \left(1 + \frac{1}{g_m R_D}\right) C_f = \left(1 + \frac{1}{g_m R_D}\right) C_f$



$$\omega_{in} = \frac{1}{R_s C_{F_1}} = \frac{1}{R_s(1+g_m R_D)C_F}$$

$$\omega_{out} = \frac{1}{R_D C_{F_2}} = \frac{1}{R_D\left(1+\frac{1}{g_m R_D}\right)C_F}$$

